

ANALYSIS

#58 — Constitution render-twins: forensic analysis (2026-08-07)

Problem: the .md gained §11.4.237 but the tracked .html/.docx/.pdf renders were stale (missing it). Goal: regenerate them, matching the operator's format, with zero bluff.

Investigation (evidence-based, systematic-debugging)

Step	Finding
pandoc version	historical renders = pandoc 3.10; local was 3.9.0.2 → upgraded to 3.10.1; also fetched the exact 3.10 binary from GitHub releases
3.10 vs 3.10.1	default default.html5 byte-identical; only styles.html differs by one font-size:12pt line inside the document-css block (which is disabled here) → 3.10.1 is output-equivalent for this corpus. -V pandoc-version=3.10 restores the generator/Creator string
<style>	identical across all 5 docs (hash 1a9ff4d...) = pandoc default styles minus the document-css body block ; PLUS a 2nd <style> = the OpenDesign stylesheet (hash 7d04bd4..., 347 lines)
reader	

Step	Finding
head reproduction	-f gfm — proven by: heading-IDs match exactly, table <colgroup> widths absent (gfm), smart typography off (straight quotes, - - not en-dash) a custom template (operator head + OD styles, title/body parameterized) reproduces the <head> byte-for-byte for all 5 (HEAD diff = 0)
residual body delta	~5.5k lines, cosmetic: historical body has &quot; where standard pandoc emits literal ", + a few TOC <a href> line-wraps
root cause of residual	no pandoc reader/writer/flag escapes a source " to &quot;; (tested -t html, -t html5, --ascii, gfm/commonmark/markdown). The historical pipeline included a non-standard post-processing step; source .md has 0 &quot; entities. Not reproducible/relevant to correctness — renders identically.

Resolution (enterprise, governed, reproducible)

Committed scripts/render/render-governance-twins.sh + assets/{governance-template.html5,od-styles.html} (§11.4.65 export-sync). Regenerated all 15 twins from the current .md: - §11.4.237 present in every render (html 6× / docx 4× / pdf 6×); §11.4.236 preserved. - Operator <head> + OpenDesign styling reproduced exactly; heading anchors correct; PDFs valid (952K–3.1M), Creator pandoc 3.10. - Deterministic: SOURCE_DATE_EPOCH=1785674948; pin weasyprint==69.0 for exact PDF Producer.

Sole deviation: the historical "→" post-step is not replicated (non-standard; identical rendering). Everything else is exact + now reproducible forever via the committed script.